FYS1105 Midtveis Prosjekt

(Trippel Pendel)

Starter med å finne Lagrange funksjonen til systemet. Først vil jeg beskrive posisjonen til de 3 pendelene ved å bruke vinklene til pendelene og lengdene mellom pendelene. Også kan jeg finne Langrange funksjonen gjennom dette. Velger i tillegg nullpunktet for potensial energi fra punktet første pendel henger.

$L = T - U$

$U = -m_1 g L_1 \cos\varphi_1 - m_2 g(L_1\cos\varphi_1 + L_2\cos\varphi_2) - m_3 g(L_1\cos\varphi_1 + L_2\cos\varphi_2 + L_3\cos\varphi_3)$

$X_1 = L_1 \sin\varphi_1$

$\dot{X}_1 = \dot{\varphi}_1 L_1 \cos\varphi_1$

$X_2 = L_1\sin\varphi_1 + L_2\sin\varphi_2$

$\dot{X}_2 = \dot{\varphi}_1 L_1\cos\varphi_1 + \dot{\varphi}_2 L_2\cos\varphi_2$

$X_3 = L_1\sin\varphi_1 + L_2\sin\varphi_2 + L_3\sin\varphi_3$

$\dot{X}_3 = \dot{\varphi}_1 L_1\cos\varphi_1 + \dot{\varphi}_2 L_2\cos\varphi_2 + \dot{\varphi}_3 L_3\sin\varphi_3$

$Y_1 = -L_1\cos\varphi_1$

$\dot{Y}_1 = \dot{\varphi}_1 L_1 \sin\varphi_1$

$Y_2 = -L_1\cos\varphi_1 - L_2\cos\varphi_2$

$\dot{Y}_2 = \dot{\varphi}_1 L_1 \sin\varphi_1 + \dot{\varphi}_2 L_2 \sin\varphi_2$

$Y_3 = -L_1\cos\varphi_1 - L_2\cos\varphi_2 - L_3\cos\varphi_3$

$\dot{Y}_3 = \dot{\varphi}_1 L_1\cos\varphi_1 + \dot{\varphi}_2 L_2\cos\varphi_2 + \dot{\varphi}_3 L_3\cos\varphi_3$

$$T = \frac{1}{2}m_1(\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2}m_2(\dot{x}_2^2 + \dot{y}_2^2) + \frac{1}{2}m_3(\dot{x}_3^2 + \dot{y}_3^2)$$

$$V = \frac{1}{2}m_1(L_1^2\dot{\varphi}_1^2) + \frac{1}{2}m_2(\dot{\varphi}_1^2L_1^2 + \dot{\varphi}_2^2L_2^2 + 2\dot{\varphi}_1\dot{\varphi}_2L_1L_2\cos(\varphi_2-\varphi_1)) + \frac{1}{2}m_3(\dot{\varphi}_1^2L_1^2 + \dot{\varphi}_2^2L_2^2 + \dot{\varphi}_3^2L_3^2 + 2(\dot{\varphi}_1\dot{\varphi}_2L_1L_2\cos(\varphi_2-\varphi_1) + \dot{\varphi}_2\dot{\varphi}_3L_2L_3\cos(\varphi_3-\varphi_2) + \dot{\varphi}_1\dot{\varphi}_3L_1L_3\cos(\varphi_3-\varphi_1)) + m_1gL_1\cos\varphi_1 + m_2g(L_1\cos\varphi_1 + L_2\cos\varphi_2) + m_3g(L_1\cos\varphi_1 + L_2\cos\varphi_2 + L_3\cos\varphi_3)$$

Videre finner jeg de deriverte, slik at jeg skal få 3 likninger senere.

$$\frac{\partial L}{\partial\dot{\varphi}_1} = m_1L_1^2\dot{\varphi}_1 + \frac{1}{2}m_2(2L_1^2\dot{\varphi}_1 + 2\dot{\varphi}_2L_1L_2\cos(\varphi_2-\varphi_1)) + \frac{1}{2}m_3(2L_1^2\dot{\varphi}_1 + 2\dot{\varphi}_2L_1L_2\cos(\varphi_2-\varphi_1) + 2\dot{\varphi}_3L_1L_3\cos(\varphi_3-\varphi_1))$$

$$\frac{\partial L}{\partial\dot{\varphi}_2} = \frac{1}{2}m_2(2L_2^2\dot{\varphi}_2 + 2\dot{\varphi}_1L_1L_2\cos(\varphi_2-\varphi_1)) + \frac{1}{2}m_3(2L_2^2\dot{\varphi}_2 + 2\dot{\varphi}_1L_1L_2\cos(\varphi_2-\varphi_1) + 2\dot{\varphi}_3L_2L_3\cos(\varphi_3-\varphi_2))$$

$$\frac{\partial L}{\partial\dot{\varphi}_3} = \frac{1}{2}m_2(2L_3^2\dot{\varphi}_3 + 2\dot{\varphi}_1L_1L_3\cos(\varphi_3-\varphi_1) + 2\dot{\varphi}_2L_2L_3\cos(\varphi_3-\varphi_2))$$

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\varphi}_1} = m_1L_1^2\ddot{\varphi}_1 + \frac{1}{2}m_2(2L_1^2\ddot{\varphi}_1 + 2L_1L_2(\ddot{\varphi}_2\cos(\varphi_2-\varphi_1) - \dot{\varphi}_2(\dot{\varphi}_2-\dot{\varphi}_1)\sin(\varphi_2-\varphi_1)) + \frac{1}{2}m_3(2L_1^2\ddot{\varphi}_1 + 2L_1L_2(\ddot{\varphi}_2\cos(\varphi_2-\varphi_1) - \dot{\varphi}_2(\dot{\varphi}_2-\dot{\varphi}_1)\sin(\varphi_2-\varphi_1)) + 2L_1L_2(\ddot{\varphi}_3\cos(\varphi_3-\varphi_1) - \dot{\varphi}_3(\dot{\varphi}_3-\dot{\varphi}_1)\sin(\varphi_3-\varphi_1)))$$

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\varphi}_2} = \frac{1}{2}m_2(2L_2^2\ddot{\varphi}_2 + 2L_1L_2(\ddot{\varphi}_1\cos(\varphi_2-\varphi_1) - \dot{\varphi}_1(\dot{\varphi}_2-\dot{\varphi}_1)\sin(\varphi_2-\varphi_1)) + \frac{1}{2}m_3(2L_2^2\ddot{\varphi}_2 + 2L_1L_2(\ddot{\varphi}_1\cos(\varphi_2-\varphi_1) - \dot{\varphi}_1(\dot{\varphi}_2-\dot{\varphi}_1)\sin(\varphi_2-\varphi_1)) + 2L_2L_3(\ddot{\varphi}_3\cos(\varphi_3-\varphi_2) - \dot{\varphi}_3(\dot{\varphi}_3-\dot{\varphi}_2)\sin(\varphi_3-\varphi_2)))$$

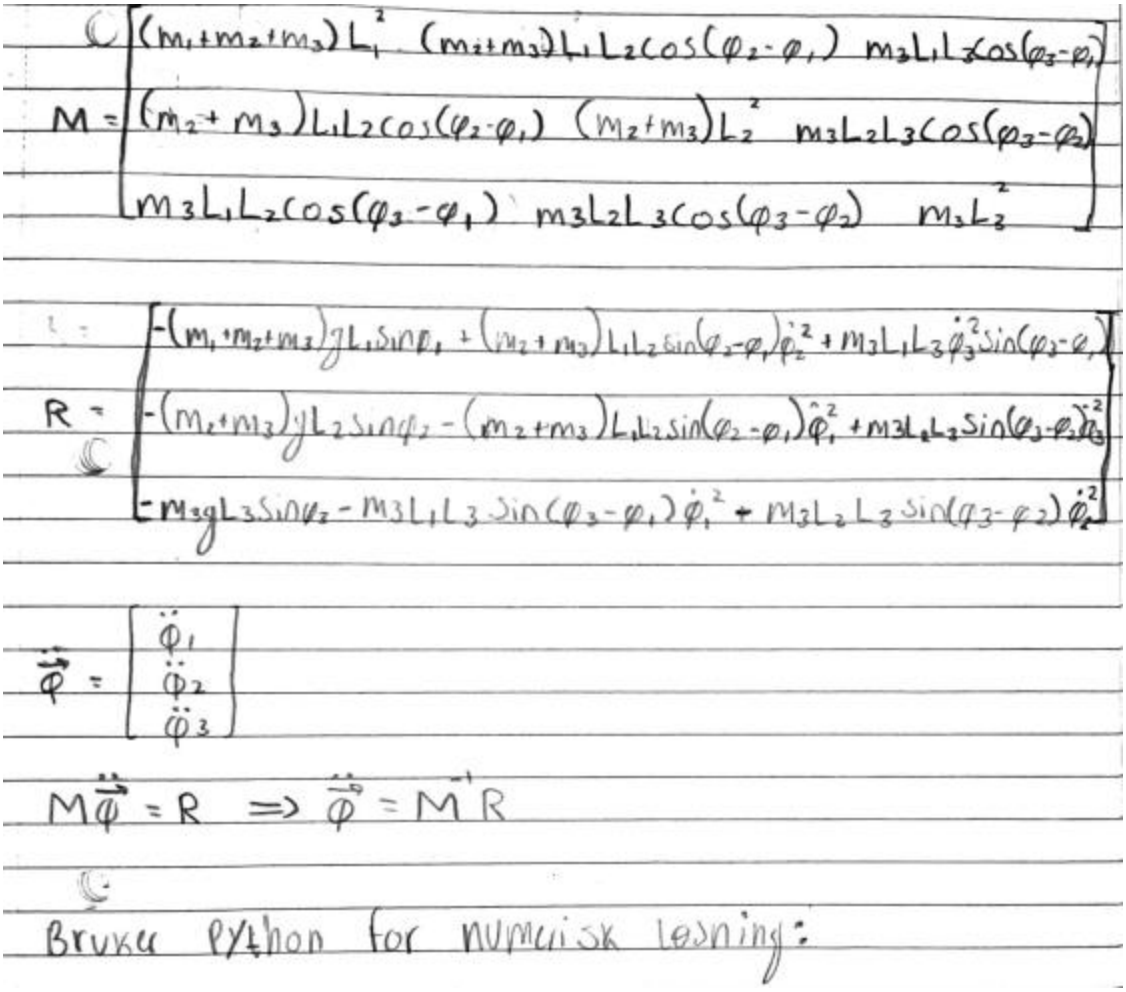
$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\varphi}_3} = \frac{1}{2}m_3(2L_3^2\ddot{\varphi}_3 + 2L_1L_3(\ddot{\varphi}_1\cos(\varphi_3-\varphi_1) - \dot{\varphi}_1(\dot{\varphi}_3-\dot{\varphi}_1)\sin(\varphi_3-\varphi_1)) + 2L_2L_3(\ddot{\varphi}_2\cos(\varphi_3-\varphi_2) - \dot{\varphi}_2(\dot{\varphi}_3-\dot{\varphi}_2)\sin(\varphi_3-\varphi_2)))$$

$$\frac{\partial L}{\partial\varphi_1} = m_2(\dot{\varphi}_1\dot{\varphi}_2L_1L_2\sin(\varphi_2-\varphi_1)) + m_3(\dot{\varphi}_1\dot{\varphi}_2L_1L_2\sin(\varphi_2-\varphi_1) + \dot{\varphi}_1\dot{\varphi}_3L_1L_3\sin(\varphi_3-\varphi_1)) - m_1gL_1\sin\varphi_1 - m_2gL_1\sin\varphi_1 - m_3gL_1\sin\varphi_1$$

$$\frac{\partial L}{\partial\varphi_2} = m_2(-\dot{\varphi}_1\dot{\varphi}_2L_1L_2\sin(\varphi_2-\varphi_1)) + m_3(-\dot{\varphi}_1\dot{\varphi}_2L_1L_2\sin(\varphi_2-\varphi_1) + \dot{\varphi}_2\dot{\varphi}_3L_2L_3\sin(\varphi_2-\varphi_3) - m_2gL_2\sin\varphi_2 - m_3gL_2\sin\varphi_2)$$

$$\frac{\partial L}{\partial\varphi_3} = m_3(-\dot{\varphi}_2\dot{\varphi}_3L_2L_3\sin(\varphi_3-\varphi_2) - \dot{\varphi}_1\dot{\varphi}_3L_1L_3\sin(\varphi_3-\varphi_1)) - m_3gL_3\sin\varphi_3$$

Etter at likningene er satt opp og forkortet kan de settes i ett matrise system som set slikt ut:



Starter med numerisk løsning:

```python
import numpy as np
from numpy.linalg import inv
import matplotlib.pyplot as plt

dt = 0.01
k = 200

#Initial verdiene

g = 9.81
m1, m2, m3 = 1, 0.5, 1
l1, l2, l3 = 1, 1, 1
p1, p2, p3 = 1, 1, 1

p1_list = np.zeros(k); p1_list[0] = p1
p2_list = np.zeros(k); p2_list[0] = p2
p3_list = np.zeros(k); p3_list[0] = p3

pd1, pd2, pd3 = 0, 0, 0

pd1_list = np.zeros(k); pd1_list[0] = pd1
pd2_list = np.zeros(k); pd2_list[0] = pd2
pd3_list = np.zeros(k); pd3_list[0] = pd3

pdd1, pdd2, pdd3 = 0, 0, 0
xaxis = np.zeros(k)
```

```
for i in range(k - 1):

    M = np.array([[(m1 + m2 + m3)*l1**2, (m2 + m3)*l1*l2*np.cos(p2 - p1), m3*l1*l3*np.cos(p3 - p1)]
        ,[(m2 + m3)*l1*l2*np.cos(p2 - p1), (m2 + m3)*l2**2, m3*l2*l3*np.cos(p3 - p2)]
        ,[m3*l1*l2*np.cos(p3 - p1), m3*l2*l3*np.cos(p3 - p2), m3*l3**2]])

    R = np.array([-(m1 + m2 + m3)*g*l1*np.sin(p1) + (m2 + m3)*l1*l2*np.sin(p2 - p1)*pd2**2 + m3*l1*l3*np.sin(p3 - p1)*pd3**2
        ,-(m2 + m3)*g*l2*np.sin(p2) - (m2 + m3)*l1*l2*np.sin(p2 - p1)*pd1**2 + m3*l2*l3*np.sin(p3 - p2)*pd3**2
        ,-m3*g*l3*np.sin(p3) - m3*l1*l3*np.sin(p3 - p1)*pd1**2 - m3*l2*l3*np.sin(p3 - p2)*pd2**2])

    #Finner den inverse til matrisen M

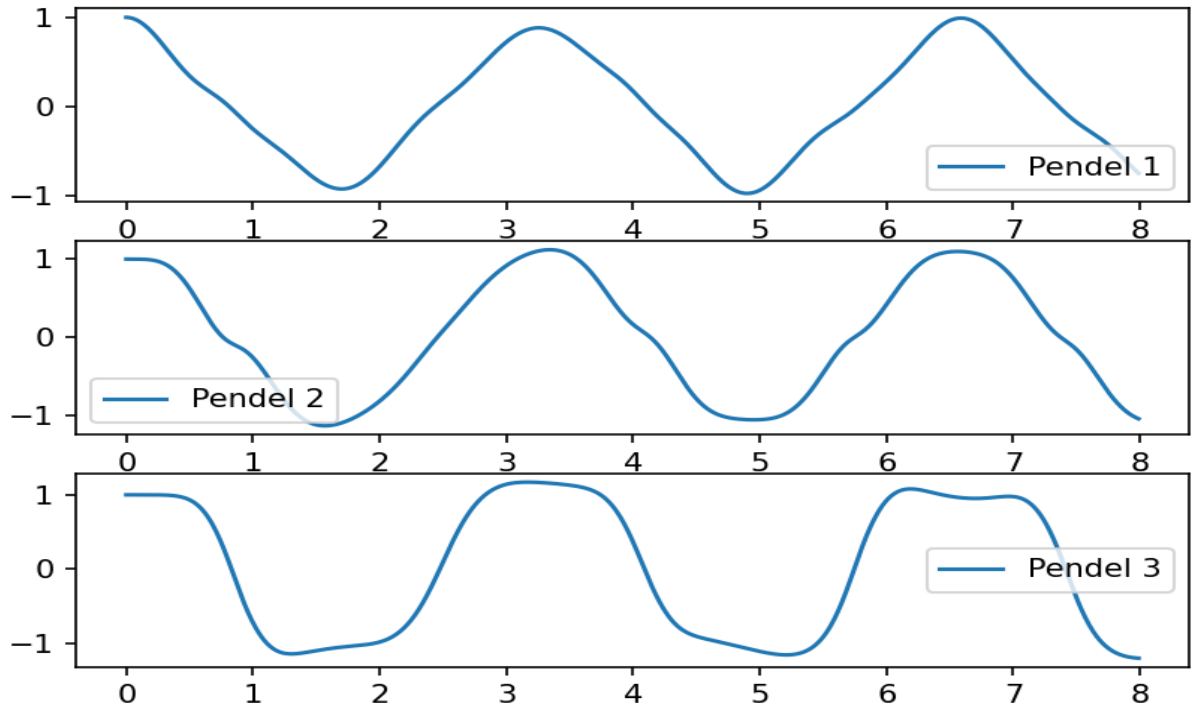
    W = inv(M)

    pdd = np.matmul(W,R)
    pdd1, pdd2, pdd3 = pdd[0], pdd[1], pdd[2]
    pd1_list[i + 1], pd2_list[i + 1], pd3_list[i + 1] = pd1_list[i] + pdd1*dt, pd2_list[i] + pdd2*dt, pd3_list[i] + pdd3*dt
    p1_list[i + 1], p2_list[i + 1], p3_list[i + 1] = p1_list[i] + pd1_list[i + 1]*dt, p2_list[i] + pd2_list[i + 1]*dt, p3_list[i] + pd3_list[
    xaxis[i + 1] = i*dt + dt

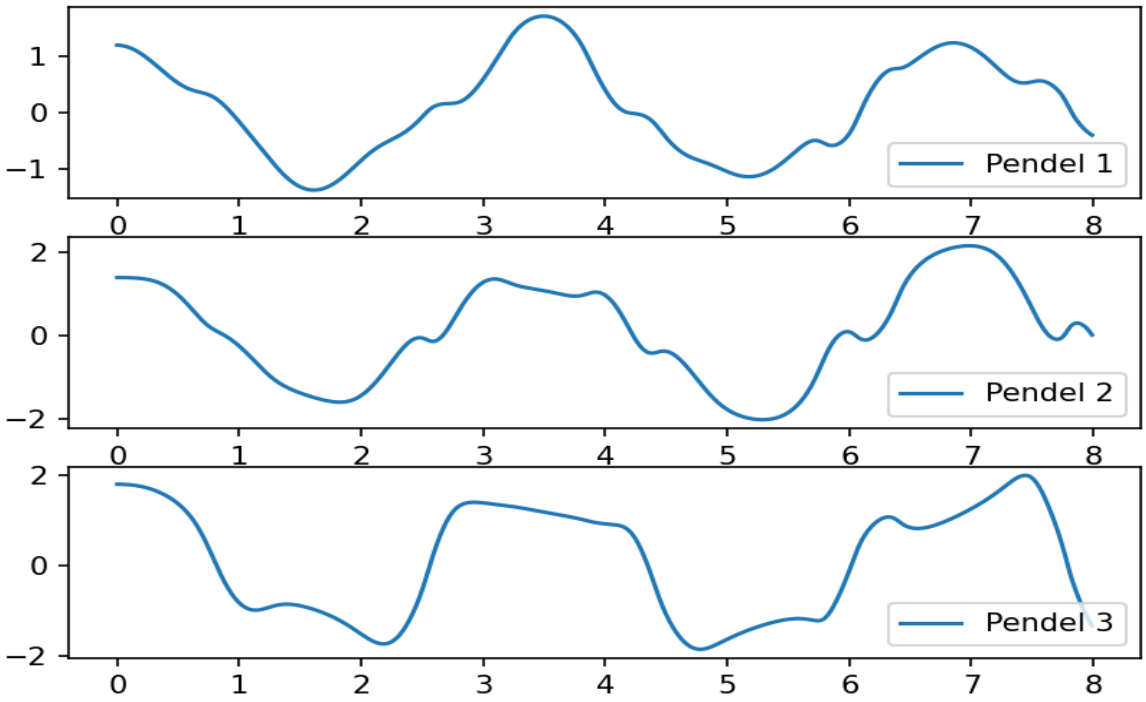
    pd1, pd2, pd3 = pd1_list[i + 1], pd2_list[i + 1], pd3_list[i + 1]
    p1, p2, p3 = p1_list[i + 1], p2_list[i + 1], p3_list[i + 1]
```

```
plt.subplot(3,1,1)
plt.plot(xaxis,p1_list,label='Pendel 1')
plt.legend()
plt.subplot(3,1,2)
plt.plot(xaxis,p2_list,label='Pendel 2')
plt.legend()
plt.subplot(3,1,3)
plt.plot(xaxis,p3_list,label='Pendel 3')
plt.legend()
plt.show()
```

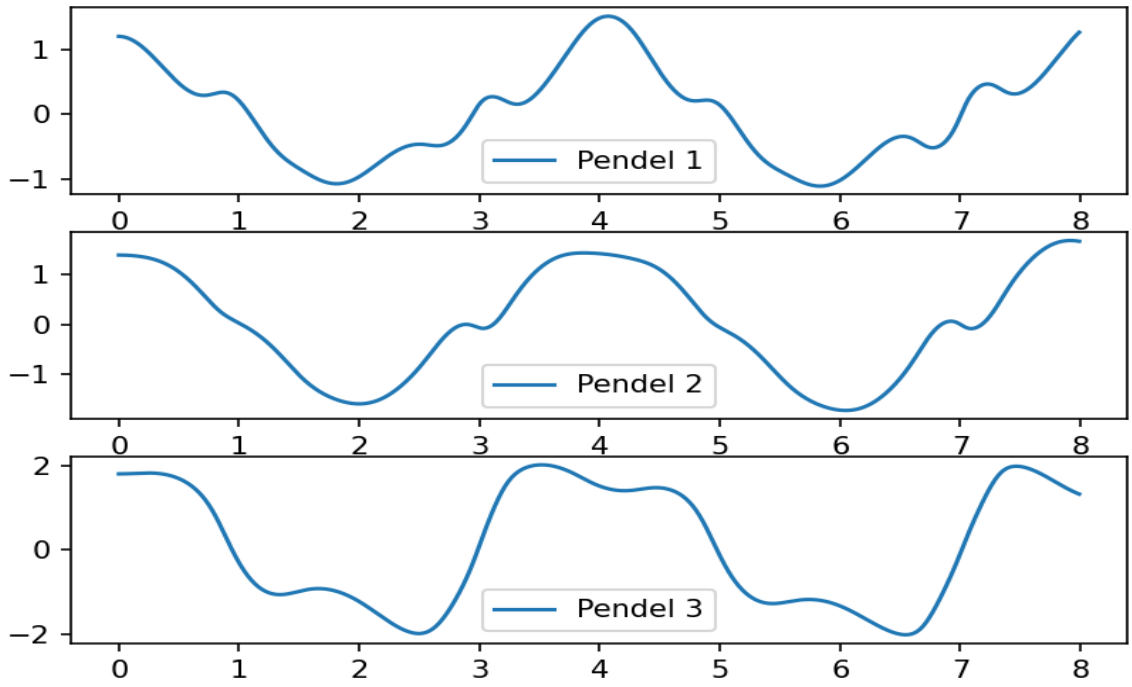
Med alle initial verdier på 1 er systemet relativt harmonisk



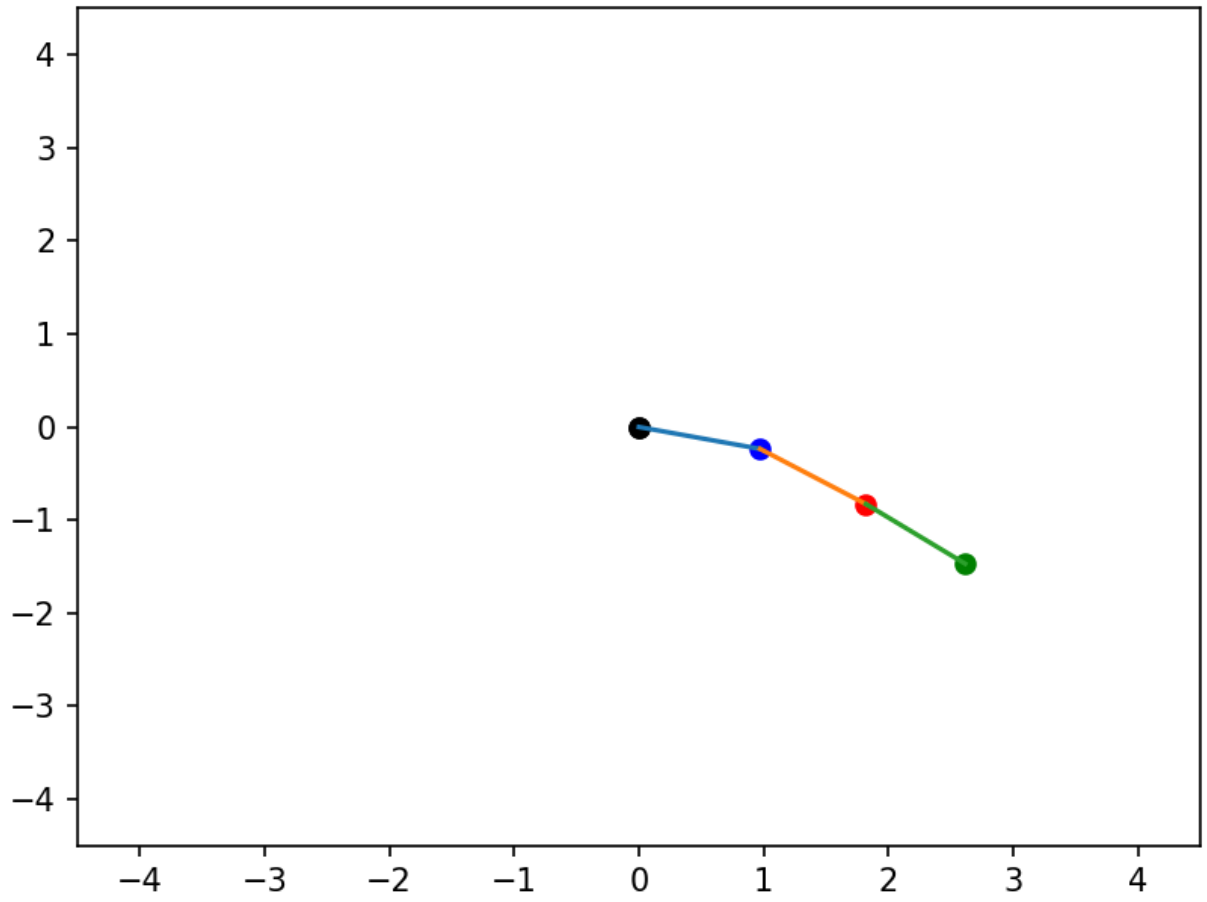
Hvis jeg endrer litt på vinkelen slik at vinklene nå er (1.2, 1.4, 1.8) istedet for (1,1,1) blir systemet mye mer kaotisk.



Hvis jeg nå i tillegg gjør at massene (fra (1,1,1) til (1.4,1.6,1.2)) er forskjelling og lengdene (fra (1,1,1) til (0.9,1.8,1.1)) er forskjelling bllir systemet enda mer kaotisk.



Lagde også en animasjon for pendelene og de tre tilfellene så ganske rimelige ut (bare et bilde fra animasjonen i tillegg til koden):




```
l = (l1 + l2 + l3)*1.5
for i in range(k):
    plt.cla()
    plt.axis([-l,l,-l,l])
    plt.plot([0,0],[0,0],'ko')
    plt.plot(np.sin(p1_list[i])*l1,-np.cos(p1_list[i])*l1, 'bo')
    plt.plot(np.sin(p2_list[i])*(l1 + l2),-np.cos(p2_list[i])*(l1 + l2), 'ro')
    plt.plot(np.sin(p3_list[i])*(l1 + l2 + l3),-np.cos(p3_list[i])*(l1 + l2 + l3), 'go')

    plt.plot([0,np.sin(p1_list[i])*l1],[0,-np.cos(p1_list[i])*l1],linestyle='-')
    plt.plot([np.sin(p1_list[i])*l1,np.sin(p2_list[i])*(l1 + l2)],[-np.cos(p1_list[i])*l1,-np.cos(p2_list[i])*(l1 + l2)],linestyle='-')
    plt.plot([np.sin(p2_list[i])*(l1 + l2),np.sin(p3_list[i])*(l1 + l2 + l3)],[-np.cos(p2_list[i])*(l1 + l2),-np.cos(p3_list[i])*(l1 + l2 + l

    plt.pause(0.001)
```

Jeg slet litt med å løse systemet analytisk selv med mange forenklinger. Jeg klarte dessverre ikke finne en generell analytisk løsning men dette er så langt jeg kom:

Forsøker å løse analytisk ved å gjøre noen forenklinger:

Vi antar at massene er lik, lengdene er like og vi har små utslag.

altså: $m_1 = m_2 = m_3 = m$,
$L_1 = L_2 = L_3 = L$,
$\sin\varphi = \varphi$ og $\cos\varphi = 1 - \frac{\varphi^2}{2}$

Da kan vi forenkle matrisene M, R og lagrangfunksjonen til:

tilnærmer også $\cos(\varphi_i - \varphi_j) \approx 1$

$$M = mL^2 \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Når vi forkorter R bruker vi også at vinklene er så små at de er tilnærma like slik at $\varphi_j - \varphi_i \approx 0$

$$R = -mgL \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Da blir ligningen:

$$M\ddot{\vec{\phi}} = R\vec{\phi}$$

så kan vi finne frekvensen ved å finne ω i:

$$\det(R - \omega^2 M) = 0$$

$$\det\left[-mgL\begin{vmatrix}3&0&0\\0&2&0\\0&0&1\end{vmatrix} - \omega^2 mL^2\begin{vmatrix}3&2&1\\2&2&1\\1&1&1\end{vmatrix}\right] = 0$$

$$\det\left[g\begin{vmatrix}3&0&0\\0&2&0\\0&0&1\end{vmatrix} + \omega^2 L\begin{vmatrix}3&2&1\\2&2&1\\1&1&1\end{vmatrix}\right]\cdot(-mL) = 0$$

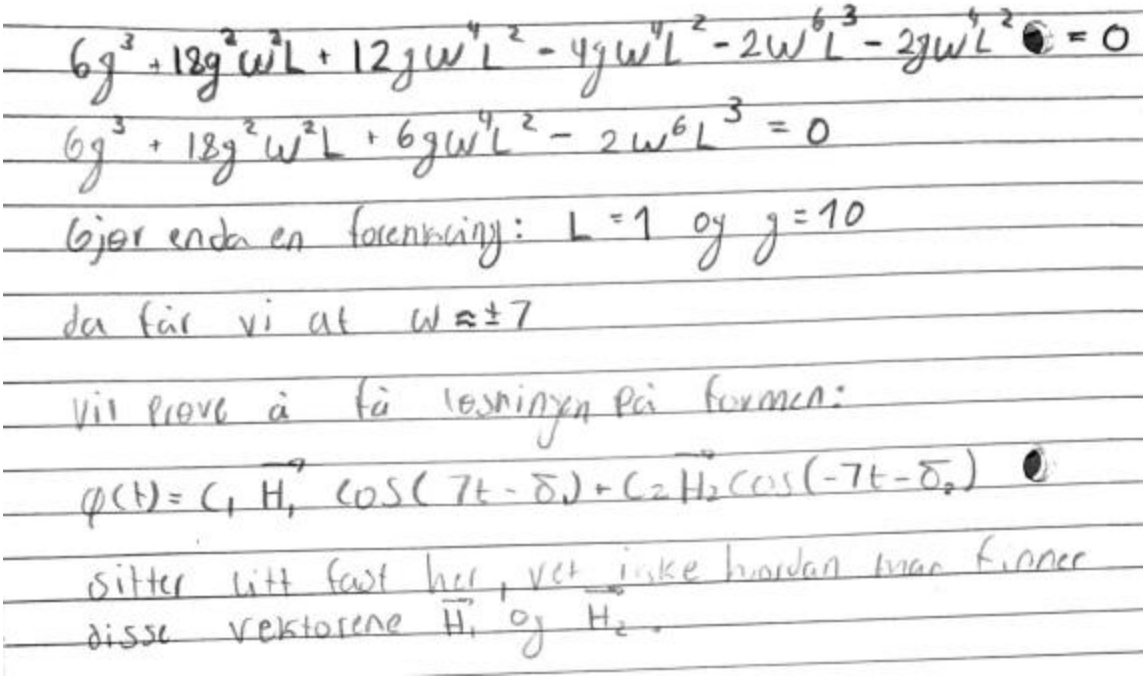
$$\begin{vmatrix}3g+3\omega^2L & 2\omega^2L & \omega^2L\\ 2\omega^2L & 2g+2\omega^2L & \omega^2L\\ \omega^2L & \omega^2L & g+\omega^2L\end{vmatrix} = 0$$

$$(3g+3\omega^2L)\begin{vmatrix}2g+2\omega^2L & \omega^2L\\ \omega^2L & g+\omega^2L\end{vmatrix} - 2\omega^2L\begin{vmatrix}2\omega^2L & \omega^2L\\ \omega^2L & g+\omega^2L\end{vmatrix} +$$

$$\omega^2L\begin{vmatrix}2\omega^2L & 2g+2\omega^2L\\ \omega^2L & \omega^2L\end{vmatrix} = 0$$

$$(3g+3\omega^2L)(2g^2+4g\omega^2L+\omega^4L^2-\omega^4L^2) - 2\omega^2L(2g\omega^2L + 2\omega^4L^2 - \omega^4L^2) + \omega^2L(2\omega^4L^2 - 2g\omega^2L - 2\omega^4L^2) = 0$$

$$(3g+3\omega^2L)(2g^2+4g\omega^2L) - 2\omega^2L(2g\omega^2L+\omega^4L^2) + \omega^2L(-2g\omega^2L) = 0$$



Tror det jeg har gjort feil er at jeg skulle finne 3 svar får omega og ikke 2. i tillegg blir det også kanskje enda et ledd i den generelle løsningen med en C3, delta 3 og H3-vektor.